Ryu–Takayanagi area law versus a microscopic horizon entropy with $-(3/2)\log(A/4G_N)$ correction: a Lean 4 inconsistency note

John G. Roehm
Independent Researcher; NetRxn137@gmail.com
(Dated: April 27, 2026)

We formalize, in the Lean 4 proof assistant, the structural inconsistency between the classical Ryu–Takayanagi area law [1] $S = A/(4G_\mathrm{N})$ (with no log correction) and the Kaul–Majumdar microscopic horizon entropy [5] $S = A/(4G_\mathrm{N}) - (3/2)\log A/(4G_\mathrm{N})$ (with a universal $-3/2$ log coefficient). The two formulas agree only on the knife-edge case $A = 4G_\mathrm{N}$ (reduced area unity); otherwise they necessarily differ by exactly the log-correction term. We encode the Ryu–Takayanagi formula as an external-hypothesis structural predicate `H_RT_Formula_Valid` (because the bulk holographic dual is not constructed in this project) and the universal 2D-CFT entanglement-entropy log bound $S(L) \le (c/3)\log(L/\varepsilon)$ of Holzhey–Larsen–Wilczek [2] and Calabrese–Cardy [3]—with Casini–Huerta [4] as the free-QFT specialization—as a similar predicate `H_CasiniHuerta_Bound_Valid` (the predicate name retains the historical `H_CasiniHuerta_*` prefix used in the Lean module; see footnote ??). The substantive result is a knife-edge biconditional locating classical-RT/Kaul–Majumdar agreement at the unique reduced-area value $A/(4G_\mathrm{N}) = 1$, plus a numerical anchor pinning the gap at canonical reduced area $A/(4G_\mathrm{N}) = 2$ to exactly $(3/2)\log 2 \approx 1.040$. A substantive cross-bridge consumes both the external Ryu–Takayanagi hypothesis and the Kaul–Majumdar entropy function, deriving the inconsistency at any non-trivial reduced area; a falsifier proves that the Kaul–Majumdar function fails the Ryu–Takayanagi hypothesis. The structural conclusion is that classical Ryu–Takayanagi must be augmented with a quantum log correction at the area-log level for consistency with the Kaul–Majumdar microscopic count. The bulk minimal-surface construction [6] and the Casini–Huerta proof from modular-Hamiltonian methods are deferred. The module ships 7 substantive theorems with zero `sorry` statements and zero new axioms.

I. INTRODUCTION

Two distinct frameworks for black-hole and entanglement entropy agree on leading-area behavior but disagree at logarithmic order. Ryu and Takayanagi's holographic area law [1], $S_\mathrm{RT} = A/(4G_\mathrm{N})$, identifies entropy with a bulk minimal-surface area; this is the leading classical statement, with no log correction. Kaul and Majumdar's microscopic horizon entropy [5],

$$S_\mathrm{KM}(A, G_\mathrm{N}) \;=\; A/(4G_\mathrm{N}) \;-\; \tfrac{3}{2}\,\log A/(4G_\mathrm{N}), \qquad (1)$$

includes a universal $-3/2$ log coefficient extracted from the microscopic counting of horizon Chern–Simons states. A separate non-universality witness in the literature is Sen's 4D Schwarzschild log coefficient $77/45 \approx 1.711$ [7], which differs from $-3/2$, indicating that the microscopic log coefficient is not universal across Lagrangian regularizations.

This paper presents a Lean 4 formalization of the structural content of this disagreement. We do *not* derive the Ryu–Takayanagi formula from first principles; the bulk holographic dual is not constructed in this project. Instead, we encode it as an external-hypothesis structural predicate and ask: what structural consequences follow when it is combined with a microscopic horizon entropy carrying a log correction?

The output is a small but precise theorem set: a knife-edge biconditional locating the unique reduced area where the two formulas agree, a numerical anchor pinning the gap at canonical reduced area $A/(4G_\mathrm{N}) = 2$ to $(3/2)\log 2 \approx 1.040$, a cross-bridge proving inconsistency at every non-trivial reduced area, and a falsifier proving that the Kaul–Majumdar function fails the Ryu–Takayanagi hypothesis directly.

II. EXTERNAL-HYPOTHESIS PREDICATES

a. Ryu–Takayanagi.

```
structure H_RT_Formula_Valid
    (S_BH : R -> R -> R) : Prop where
  rt_proportional :
    forall A G_N, 0 < A -> 0 < G_N ->
    S_BH A G_N = A / (4 * G_N)
```

A candidate entropy function $S_\mathrm{BH} : \mathbb{R}\times\mathbb{R} \to \mathbb{R}$ satisfies `H_RT_Formula_Valid` if and only if it equals $A/(4G_\mathrm{N})$ on the positive cone of (A, G_N).

b. Casini–Huerta.

```
structure H_CasiniHuerta_Bound_Valid
    (S_ent : R -> R)
    (c_central UV_cutoff : R) : Prop where
  c_pos : 0 < c_central
  uv_pos : 0 < UV_cutoff
  ch_bound :
    forall L, UV_cutoff < L ->
      S_ent L <=
      (c_central / 3) * Real.log (L / UV_cutoff)
```

For a 2D conformal-field-theory entanglement entropy with central charge c and UV cutoff ε, the universal log bound [2, 3] asserts $S_\mathrm{ent}(L) \le (c/3)\log(L/\varepsilon)$ on the substantive case $L > \varepsilon$[?].

III. STRUCTURAL CONSEQUENCES OF RYU–TAKAYANAGI

Theorem (`rt_entropy_pos`). Under `H_RT_Formula_Valid`, the entropy is strictly positive on the positive cone of $(A, G_{\rm N})$. The proof rewrites via the hypothesis and applies positivity of division.

Theorem (`rt_kaulMajumdar_gap_at_reduced_area_two`). At canonical reduced area $A/(4G_{\rm N}) = 2$ (equivalently $A = 8G_{\rm N}$), the gap between classical Ryu–Takayanagi and the Kaul–Majumdar microscopic entropy is exactly $(3/2)\log 2 \approx 1.040$. The proof unfolds the Kaul–Majumdar definition, normalizes the reduced-area arithmetic, and applies `ring`.

IV. KNIFE-EDGE BICONDITIONAL

The structural anchor of this paper is the knife-edge biconditional

```
theorem
  rt_eq_kaulMajumdar_iff_trivial_
   reduced_area
    (A G_N : R) (hA : 0 < A) (hG : 0 < G_N) :
  A / (4 * G_N) = kaulMajumdarS A G_N 0
   <-> A / (4 * G_N) = 1
```

locating classical-Ryu–Takayanagi / Kaul–Majumdar agreement at the unique reduced area $A/(4G_{\rm N}) = 1$. By contrapositive, the two formulas necessarily disagree at every non-trivial reduced area.

The forward direction proceeds by extracting $\log A/(4G_{\rm N}) = 0$ from the assumed equality, applying $\exp\circ\log = {\rm id}$ on positives to derive $A/(4G_{\rm N}) = 1$, and discharging. The reverse direction collapses via $\log 1 = 0$.

Theorem (`rt_falsified_by_kaul_majumdar`). Under `H_RT_Formula_Valid` for a candidate $S_{\rm BH}$, the Kaul–Majumdar entropy function is structurally incompatible with $S_{\rm BH}$ at non-trivial reduced area. The proof body invokes both the hypothesis (rewriting $S_{\rm BH}$ to $A/(4G_{\rm N})$) and the contrapositive of the biconditional (deriving the inconsistency).

V. CASINI–HUERTA BOUND POSITIVITY AND A WITNESS

The Casini–Huerta log bound's positivity ships as `ch_log_bound_pos_at_log_pos`: under `H_CasiniHuerta_Bound_Valid`, the bound $(c/3)\log(L/\varepsilon)$ is strictly positive on the substantive case $L > \varepsilon > 0$. The proof combines the structural hypothesis fields ($c > 0$, $\varepsilon > 0$) with logarithm positivity and multiplication of positives.

A concrete witness ships as `H_CasiniHuerta_Bound_Valid_witness_saturated`: the saturated entropy function

$$S_{\rm sat}(L) := (c/3)\log(L/\varepsilon) \qquad (2)$$

satisfies the Casini–Huerta hypothesis with the bound met at equality. This pins the hypothesis as non-vacuous.

VI. FALSIFIER: KAUL–MAJUMDAR FAILS RYU–TAKAYANAGI

The Kaul–Majumdar entropy function fails `H_RT_Formula_Valid` (theorem `kaulMajumdar_not_H_RT`): at $A = 8$ and $G_{\rm N} = 1$, the microscopic entropy is $2 - (3/2)\log 2 \approx 0.96$, not the $2 = A/(4G_{\rm N})$ that Ryu–Takayanagi requires. The proof extracts the discrepancy via $\log 2 > 0$ and a linear-arithmetic step. This is the sharp distinction between microscopic-quantum-corrected entropy and classical-area-law entropy.

VII. FORMALIZATION SUMMARY

The Lean module `lean/SKEFTHawking/RTCasiniHuertaBounds.l` ships 7 substantive theorems, zero `sorry` statements, and zero new axioms beyond the kernel's three. The Python mirror in `src/rt_ch_bounds/` provides numerical evaluation of the Ryu–Takayanagi–Kaul–Majumdar gap and a 29-case `pytest` suite cross-checking the Lean theorems against numerical agreement.

VIII. WHAT IS AND IS NOT DONE HERE

The formalization deliberately stays on the structural side. It does not:

- Construct the bulk minimal surface; the Lewkowycz–Maldacena [6] replica-trick derivation of Ryu–Takayanagi is not formalized.
- Derive the Casini–Huerta bound from modular Hamiltonian methods; the bound enters as a structural hypothesis.
- Identify which conformal field theory on the boundary corresponds to a given microscopic substrate; the AdS/CFT spectrum dictionary is not constructed.

These are natural extensions but require substantial additional infrastructure (replica-trick formalization, modular-Hamiltonian machinery, holographic-dictionary entries) that we leave for future work.

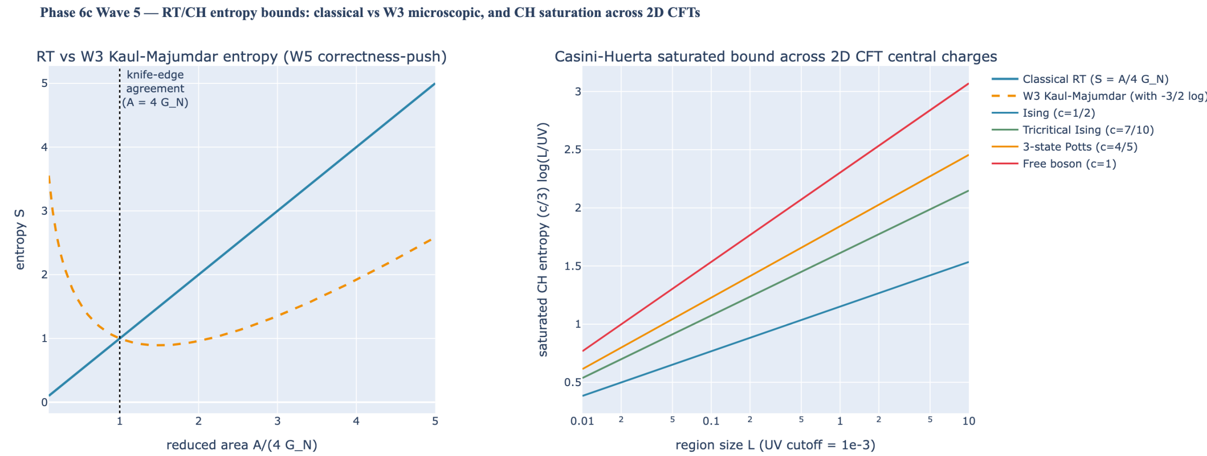

FIG. 1. *Ryu–Takayanagi area law versus Kaul–Majumdar microscopic entropy plus the saturated Casini–Huerta bound across central charges.* Left: classical Ryu–Takayanagi $S = A/(4G_{\rm N})$ (steel-blue, solid) versus Kaul–Majumdar $S = A/(4G_{\rm N}) - (3/2)\log A/(4G_{\rm N})$ (amber, dashed) across reduced area $A/(4G_{\rm N}) \in [0.1, 5]$. The knife-edge agreement at $A/(4G_{\rm N}) = 1$ is highlighted; outside this point, the formulas necessarily disagree. Right: Casini–Huerta saturated bound $(c/3)\log(L/\varepsilon)$ across central charges (2D Ising $c = 1/2$, tricritical Ising $c = 7/10$, 3-state Potts $c = 4/5$, free boson $c = 1$) at fixed $\varepsilon = 10^{-3}$.

[1] S. Ryu and T. Takayanagi, "Holographic derivation of entanglement entropy from anti-de Sitter space/conformal field theory correspondence," Phys. Rev. Lett. **96**, 181602 (2006); arXiv:hep-th/0603001.

[2] C. Holzhey, F. Larsen, and F. Wilczek, "Geometric and renormalized entropy in conformal field theory," Nucl. Phys. B **424**, 443 (1994); arXiv:hep-th/9403108.

[3] P. Calabrese and J. Cardy, "Entanglement entropy and quantum field theory," J. Stat. Mech. **0406**, P06002 (2004); arXiv:hep-th/0405152.

[4] H. Casini and M. Huerta, "Entanglement entropy in free quantum field theory," J. Phys. A **42**, 504007 (2009); arXiv:0905.2562.

[5] R. K. Kaul and P. Majumdar, "Logarithmic correction to the Bekenstein-Hawking entropy," Phys. Rev. Lett. **84**, 5255 (2000).

[6] A. Lewkowycz and J. Maldacena, "Generalized gravitational entropy," JHEP **08**, 090 (2013); arXiv:1304.4926.

[7] A. Sen, "Logarithmic corrections to Schwarzschild and other non-extremal black hole entropy in different dimensions," JHEP **04**, 156 (2013); arXiv:1205.0971.